

LESSON 3-4 • SESSION 2 • PRACTICE • FORM B

Interpreting Points on Ratio Graphs

Learning target – I can explain what a point on a ratio graph means in the situation it describes. • 6.RP.A.3a

REMEMBER

- A point (x, y) says x of one thing goes with y of another. **Read the axis labels.**
- The **unit rate point** is $(1, r)$. If x is not 1, divide: for $(3, 15)$, $15 \div 3 = 5$, so $(1, 5)$.
- To find x from y , divide y by the unit rate: rate 5, $y = 30$, so $x = 6$.

WORKED EXAMPLE

x = pounds of apples, y = cost in dollars. The line passes through $(4, 12)$. Find the cost of 6 pounds.

Step 1 Read it: 4 pounds cost \$12.

Step 2 Divide: $12 \div 4 = 3$, so the unit rate point is $(1, 3)$.

Step 3 Multiply: $6 \times 3 = 18$, so $(6, 18)$.

Answer \$3 per pound; 6 pounds cost \$18

WORDS unit rate point (*punto de tasa unitaria*) • interpret (*interpretar*) • coordinates (*coordenadas*)

PRACTICE

show your work • label units

1. Find the unit rate point $(1, r)$ for a line through each point.

a. $(2, 14)$ _____

b. $(5, 30)$ _____

c. $(4, 6)$ _____

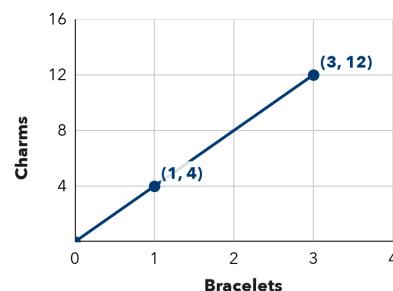
d. $(3, 27)$ _____

2. **Jalen's Charms.** The graph shows bracelets (x) and charms (y).

a. What does $(0, 0)$ mean? _____

b. What does $(1, 4)$ mean? _____

c. Charms for 7 bracelets: _____ charms



3. **Water Pump.** A graph of minutes (x) and gallons (y) passes through $(1, 6)$ and $(5, 30)$.

Hint Divide the gallons by the unit rate.

a. Unit rate: _____ gallons per minute

b. What does $(5, 30)$ mean? _____

c. Minutes to pump 48 gallons: _____ minutes

4. **Snow Shoveling.** A graph of driveways (x) and dollars (y) passes through $(1, 12)$ and $(4, 48)$. What is the pay for 7 driveways?

Hint Use the unit rate point $(1, 12)$.

Show your work.

Answer _____ dollars

5. **Drone Flight.** A graph of seconds (x) and meters (y) passes through $(1, 11)$. What point shows 6 seconds?

Show your work.

Point _____

6. **Robot Speed.** A graph of seconds (x) and meters (y) passes through (3, 15).

Hint Divide 15 by 3 to get the unit rate.

- a. Unit rate point: _____
- b. Distance in 9 seconds: _____ m

Show your work.

7. **Bike Racks.** A graph of racks (x) and spaces (y) passes through (3, 18). Find the unit rate point. Is (8, 48) on the line?

Hint Check: does $8 \times$ the unit rate equal 48?

Show your work.

Yes or no _____

8. **Two Bike Riders.** Both graphs show hours (x) and kilometers (y). Rider 1’s line passes through (2, 30). Rider 2’s passes through (3, 36).

- a. Each rider’s unit rate: _____
- b. Whose line is steeper? What does that mean? _____
- c. How far ahead is the faster rider after 5 hours? _____ km

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Priya’s Work.** A graph of tickets (x) and cost in dollars (y) passes through (1, 6). Priya says, “This means 6 tickets cost 1 dollar.” Explain her mistake and say what the point really means.

TEST PRACTICE

choose the best answer

10. A point (3, 24) on a graph of boxes (x) and pens (y) means –
- A 24 boxes hold 3 pens

B 3 boxes hold 24 pens

C each box holds 3 pens

D there are 27 pens in all

11. A ratio graph passes through (6, 42). The unit rate is –
- A 36

B 48

C 7

D 252

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

288 Z	11 X	(6, 66) T	15 D	45 E	8 A	28 H	(1, 1.5) C	84 R	(66, 6) Q
#1c		#2c	#3c	#4	#5	#6b	#8c		

CHECK YOURSELF

☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____